

of $\mathbf{x}$ are related to the first factor; the next three columns are related to the first factor; and the final column is an interaction term between the two factors. Both the DGM and skip-DGM should be able to reconstruct the data using $K = 2$ dimensions, whereas PPCA would need at least $K = 5$ latent dimensions to capture the nonlinear terms.

For the DGM and skip-DGM, we use a similar neural network architecture as the previous section, except with 50 neurons in each hidden layer. For PPCA, we consider both a model with the true number of latent dimensions, $K = 2$, and a model with an overestimate of the latent dimension, $K = 5$.

We first check each model. PPCA with $K = 2$ fails the partial predictive check, as expected; two-dimensions are inadequate for a linear model to capture the data. Each of PPCA ($K = 5$), the DGM and skip-DGM pass their partial predictive check (Figure 7). To assess which model (or set of models) to proceed with, we use the PPN study. Consider Figure 7:

- (row 2, column 3): The PPN comparing PPCA-5 and the DGM (with the PPCA-5 diagnostic) suggests that the DGM can fool PPCA-5.
- (row 2, column 3): The PPN comparing PPCA-5 and the skip-DGM (with the PPCA-5 diagnostic) suggests that the DGM can fool PPCA-5.
- (row 3, column 2): The PPN comparing PPCA-5 and the DGM (with the DGM diagnostic) suggests that PPCA-5 cannot fool the DGM.
- (row 3, column 4): The PPN comparing the DGM and the skip-DGM (with the DGM diagnostic) suggests that the skip-DGM can fool the DGM.
- (row 4, column 2): The PPN comparing PPCA-5 and the skip-DGM (with the skip-DGM diagnostic) suggests that the skip-DGM can fool PPCA-5.
- (row 4, column 3): The PPN comparing the DGM and the skip-DGM (with the skip-DGM diagnostic) suggests that the DGM can fool the skip-DGM.

The skip-DGM fools PPCA-5, but PPCA-5 does not fool the skip-DGM. This result suggests that the skip-DGM captures aspects of the data that PPCA-5 does not. Based on the overlap of the posterior predictive distributions, both the DGM and skip-DGM fool each other, suggesting that they both capture the same aspects of the data.

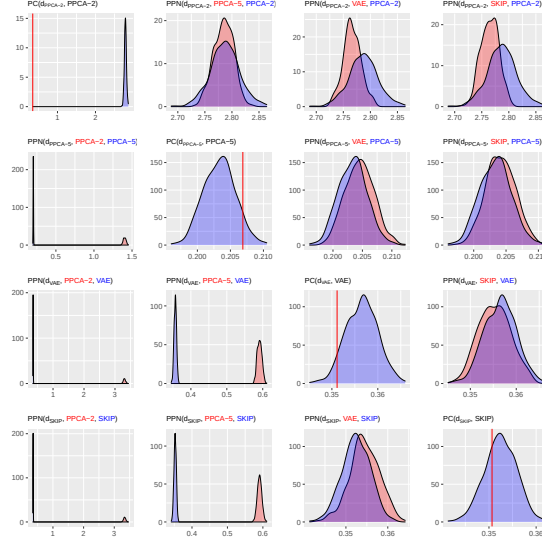


Figure 7: A PPN study of factor analysis models. This study suggests that a nonlinear model is consistent with the data (a linear model is not enough). On the diagonal are heldout predictive checks displaying $p_A(d_A(\mathbf{x}_{\text{rep}}^A; \mathbf{x}_{\text{val}})|\mathbf{x}_{\text{in}})$ (blue histogram) and $d_A(\mathbf{x}_{\text{out}}; \mathbf{x}_{\text{val}})$ (red line). PPCA-5, VAE and the skip-VAE pass their model checks. Beside the diagonal are PPNs, each one checking if a different model can fool the model under study. Specifically, the PPNs compare $p_B(d_A(\mathbf{x}_{\text{rep}}^B; \mathbf{x}_{\text{val}})|\mathbf{x}_{\text{in}})$ (red histogram) and $p_A(d_A(\mathbf{x}_{\text{rep}}^A; \mathbf{x}_{\text{val}})|\mathbf{x}_{\text{in}})$ (blue histogram). The PPN study shows that PPCA-5 does not fool the skip-VAE and VAE, while the skip-VAE and VAE fool each other.

Table 3:: Symmetrized Kullback-Leibler distance between distributions for the PPN study in Figure 7. This study suggests that a nonlinear model is consistent with the data.

	PPCA-2	PPCA-5	VAE	SKIP-VAE
PPCA-2		0.43	1.59	1.72
PPCA-5	12.44		0.37	0.26
VAE	12.68	14.31		0.26
SKIP-VAE	12.21	14.31	0.47	

4 Discussion

We developed and studied the posterior predictive null check (PPN), an approach to Bayesian model criticism that complements the classical predictive checks. A PPN checks whether data from model B's posterior predictive distribution can pass the predictive check of model A. By studying a space of models with a collection of PPNs, we can understand the relationships between them. Which models capture different aspects of the data?

With mixtures, we demonstrated how a PPN study can help select a model by the principle of parsimony. With probabilistic factor models, we demonstrated how it can

help understand relationships between different classes of models. We re-analyzed data from the research literature on Bayesian model criticism, and we studied the calibration properties of the PPN.

Running a PPN study is more computationally expensive than computing predictive checks. This expense is because for M models, a PPN study considers order M^2 model combinations. This computational expense may be mitigated when the models are ordered by complexity. In this case, a PPN study can proceed by comparing only consecutive models (i.e. $\mathcal{M}_k$ vs. $\mathcal{M}_{k+1}$), reducing the number of model combinations to M .

In the modern practice of applied Bayesian statistics, researchers iteratively design and explore many models, a process that was recently dubbed “the Bayesian workflow” (Gelman et al., 2020). By helping the researcher understand the relationships between different models, and particularly so in the context of Bayesian model criticism, a PPN study can help guide the researcher through this process.

All of the PPN studies here, both with real data and simulated data, involve a situation where more than one model passes its predictive check. One interesting direction of future work is to consider a PPN study where no model passes its check, but where we might still be interested in understanding the differences between the models’ predictive distributions.

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